

Physics-Informed Machine Learning for Predicting Supporting-Boundary Circularity at Any Tool Position in Micro-EDM of Metallic Lattices

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Research Paper — Ideation, Data Expansion, Training/Validation, and Position-Aware Prediction

Abstract

We present a physics-informed machine-learning method that predicts the circularity ratio of the supporting material boundary after micro-electrical discharge machining (EDM) of a metallic lattice, for **any tool landing position** (x, y) and any EDM recipe (peak current I , pulse-on time T , duty factor D). Only sixteen laboratory runs with SEM labels were available. From these, we (i) generated 1,100 synthetic EDM points via a Gaussian Process posterior over the process inputs, and (ii) expanded each labeled run across a spatial landing grid by modulating SEM circularity with a geometry-risk penalty, producing hundreds of position-aware training rows. Dual Gradient Boosting regressors learn circularity (1–5) and supporting integrity; predictions are blended with EDM physics heuristics. Circularity ratio is reported as score/5. Leave-one-out and cross-validation guide trust. SEM evidence shows only Run 4 (4 A, 150 μ s, 80%) yields an intact circular supporting ring — the objective the models are trained to maximize.

Keywords: micro-EDM; lattice structures; circularity ratio; Gaussian process; gradient boosting; physics-informed ML; data augmentation; SEM validation

1. Problem Definition

A metallic lattice has square unit cells of side 500 μ m. Pore and node diameters equal $x = 500\sqrt{2} / 3 \approx 235.6$ μ m. The EDM tool tip is 900 μ m (tool/pore = 3.82), so the electrode always overlaps pores, nodes, and supporting struts together. Success means a nearly circular open region with a **continuous supporting ring**; nodes may be destroyed. Hole-deviation numbers alone are insufficient: among sixteen SEM-validated trials, only Run 4 produced the desired ring.

The scientific question is therefore: given (I, T, D, x, y) , predict circularity ratio and whether the supporting boundary survives — even at positions never machined in the lab.

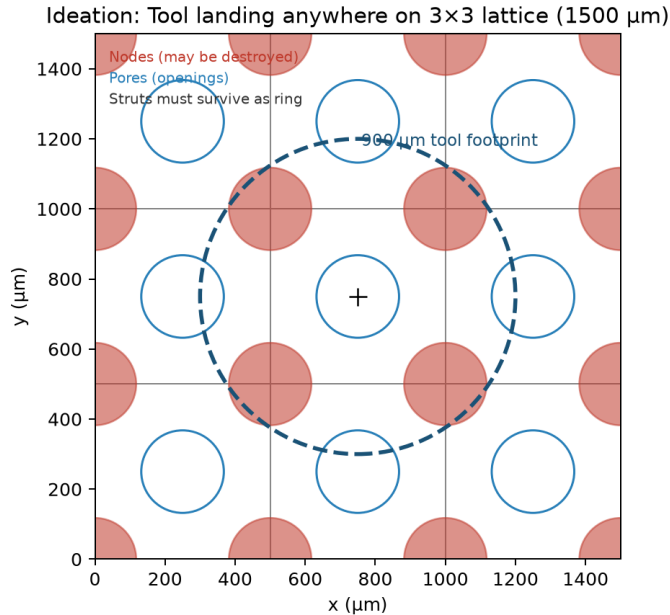


Fig. 1. Ideation schematic: 900 μ m tool footprint on a 1500×1500 μ m (3×3 cell) working area. Any landing position changes strut/node overlap and therefore circularity risk.

2. Ideation — The Working Mind of the Solution

The core idea is that circularity is not a property of EDM parameters alone. It is the **joint effect of process intensity and local lattice geometry under the tool**. Two landings with identical (I , T , D) can succeed at a pore center and fail near a strut because geometry risk changes. Conversely, even a favorable landing fails if current is too aggressive. That coupling is the “soul” of the method.

2.1 Three ideation principles

Principle 1 — Objective from SEM, not metrology proxies. Train to maximize supporting-boundary circularity judged from SEM, not to minimize Hole_Dev_Top/Bottom. Run 5 looks “best” by deviation but destroys the ring; Run 4 is the only SEM success.

Principle 2 — Geometry is a first-class feature. Before any ML call, a lattice geometry engine computes how the tool footprint intersects nodes, pores, and struts at (x , y). Those quantities enter the feature vector beside EDM descriptors.

Principle 3 — Sparse lab truth + physics-structured expansion. Sixteen runs cannot cover every (recipe $\times$ position). Expand data by (a) GP sampling in EDM space and (b) replaying each SEM label across a landing grid with geometry penalties that encode the physical belief “higher strut risk $\rightarrow$ lower circularity.”

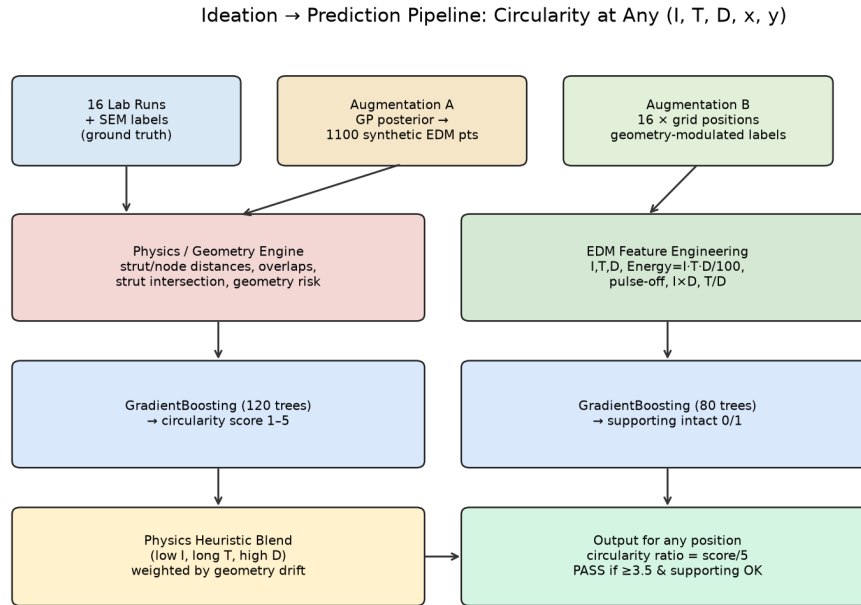


Fig. 2. End-to-end ideation pipeline: sparse SEM truth $\rightarrow$ dual augmentation $\rightarrow$ physics features + EDM features $\rightarrow$ Gradient Boosting + heuristic blend $\rightarrow$ circularity ratio at any position.

3. Physics Used in the Predictor

3.1 EDM process quantities

From machine settings we compute discharge energy and related intensity terms:

$$E = I \cdot T \cdot (D/100), \quad \text{pulse-off} \approx T \cdot (100 - D)/D, \quad I \times D, \quad T/D.$$

Physically, low current keeps a small plasma channel; long pulse-on spreads erosion more uniformly in the radial direction; high duty sustains continuous removal. Heuristics therefore reward $I \leq 5$ A, $T \geq 130$ μ s, $D \geq 75\%$, and penalize $I \geq 8$ A (strut blasting).

3.2 Lattice geometry quantities at a landing

For tool center (x, y) on a $1500\ \mu\text{m}$ working area, the engine evaluates:

- minimum distance to nearest strut and nearest node center;
- counts of nodes/pores inside the tool footprint;
- strut intersection length inside the tool;
- pore and node overlap fractions;
- tool/pore ratio and a scalar **geometry risk** index:

$$\text{geometry_risk} = 0.5 \cdot \text{strut_risk} + 0.3 \cdot \text{intersect_risk} + 0.2 \cdot \text{ratio_factor} \in [0,1].$$

High geometry risk means the oversized tool is heavily chewing supporting ligaments — circularity is expected to fall even if the EDM recipe is gentle.

4. From 16 Experiments to a Trainable Dataset

Ground truth consists of sixteen EDM trials labeled from SEM on a 1–5 boundary-circularity scale plus a binary supporting-intact flag. Only Run 4 scores 5 with an intact ring.

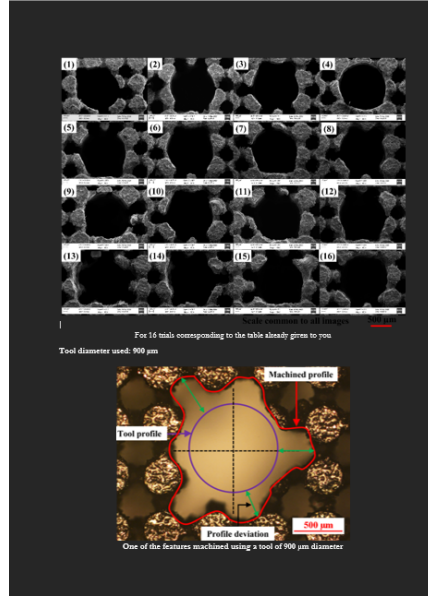


Fig. 3. SEM montage of the sixteen experimental outcomes used as visual ground truth.

Run	I	T	D	Circ	Intact	Role
4*	4	150	80	5	Yes	SEM success / target
5	6	50	64	2	No	Best deviation, FAIL SEM
1	4	50	56	1	No	Low-energy fail
8	6	150	72	2	No	Asymmetric boundary
13	10	50	80	1	No	High-current blast
16	10	150	56	1	No	Irregular despite low Dev↓

Table 1. Representative labeled runs (* = only SEM success).

4.1 Augmentation A — Gaussian Process synthetic EDM points (1,100)

A Gaussian Process was fit on the sixteen real $(I, T, D) \rightarrow$ response mappings (hole deviations / related process outputs). Sampling the GP posterior inside the experimental bounds generated **1,100 synthetic points** (file `synthetic_1100_points.csv`, source tag `GP_posterior`). These densify EDM-parameter space for exploratory Phase 1 models. They are **not** new laboratory experiments; final SEM-anchored recommendations still privilege the sixteen real labels.

4.2 Augmentation B — Spatial expansion of SEM labels (position-aware rows)

To predict circularity at arbitrary (x, y) , each of the 16 SEM labels is replayed on a discrete landing grid inside the valid tool-center bounds (step $\approx 150 \mu\text{m} \rightarrow 25$ landings $\rightarrow \approx 400$ training rows). For landing geometry features g and SEM circularity c_{SEM} of that run:

$$c_{\text{train}} = \text{clip}(c_{\text{SEM}} - 2.5 \cdot \text{geometry_risk}(g) + \text{edm_bonus}(I, T, D), 1, 5).$$

Here *edm_bonus* is +0.5 for gentle recipes ($I \leq 5, T \geq 130, D \geq 75$), -1.0 for $I \geq 8$, else 0. Supporting-integrity targets are set to 0 when current is high or geometry risk is large. This encodes the ideation that the same recipe degrades when the tool sits on dense strut intersections.

How 16 Experiments Become Thousands of Training Rows

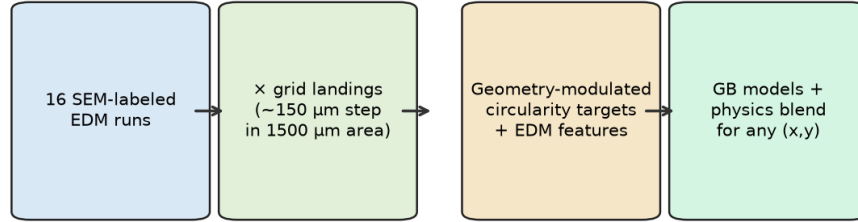


Fig. 4. Data-expansion path: 16 SEM-labeled runs $\rightarrow$ grid landings $\rightarrow$ geometry-modulated targets $\rightarrow$ models that generalize to unseen positions.

5. Machine Learning Models, Training, Testing, Validation

5.1 Feature vector (20 dimensions)

EDM block (7): I, T, D , energy E , pulse-off proxy, $I \times D, T/D$. **Geometry block (13):** normalized (x, y) , min distances to strut/node, nodes/pores inside tool, strut intersection length, pore/node overlap fractions, geometry risk, tool/pore ratio, working area, tool diameter.

5.2 Model architecture

Two supervised Gradient Boosting regressors (scikit-learn):

- Circularity model: 120 estimators, max depth 4 $\rightarrow$ score $\in [1, 5]$.
- Supporting-integrity model: 80 estimators, max depth 3 $\rightarrow$ continuous score thresholded at 0.5.

Phase 1 (position-unknown) also uses a Gaussian Process Regressor with Matérn 5/2 + WhiteKernel on the 16 SEM circularity labels to search favorable (I, T, D) neighborhoods, and a polynomial Ridge model under leave-one-out cross-validation (LOOCV).

5.3 Train / test / validate protocol

Training set (position-aware): geometry-expanded rows from all 16 runs (≈ 400 samples at $150 \mu\text{m}$ grid). Models are fit on the full expanded matrix and persisted.

Validation (small-n honest checks): LOOCV on the 16 real SEM circularity labels for Phase 1 Ridge/GP analysis (MAE on the order of ~ 1 point on the 1–5 scale). For exploratory 16+synthetic models, 5-fold cross-validated MAE on hole-deviation responses monitors overfitting in EDM space. Because the labeled set is tiny and unbalanced (one clear success), validation metrics are decision-support, not proof of industrial accuracy — SEM remains the ultimate test.

Physics blend at inference (test-time regularization): final circularity = $(1-w) \cdot \text{ML} + w \cdot \text{heuristic}$, where weight w grows when tool/pore sizes drift from the lab geometry (900 μm / 235.6 μm). This prevents pure ML hallucination far from the training manifold.

Stage	Data	Method	Purpose
Phase 1 fit	16 SEM labels	GP (Matérn 5/2)	Search robust (I,T,D)
Phase 1 CV	16 runs LOOCV	Polynomial Ridge	Estimate circularity MAE
Synthetic densify	GP posterior	1,100 points	Fill EDM input space
Spatial expand	16 $\times$ ~25 landings	Risk-modulated labels	Teach position effect
Position model	~400 rows / 20 feats	GBR 120 + GBR 80	Circularity & support
Inference	Any (I,T,D,x,y)	ML + heuristic blend	Ratio = score/5

Table 2. Training, validation, and inference stack.

6. Predicting Circularity Ratio for Any Position

Given a query (I, T, D, x, y) :

1. Run the geometry engine at $(x, y) \rightarrow$ risk and overlap features.
2. Build the 20-D feature row (EDM + geometry).
3. Predict ML circularity score $c_{\text{ML}} \in [1,5]$ and supporting flag.
4. Evaluate physics heuristic c_{H} from the gentle/aggressive rules and risk.
5. Blend: $c = (1-w) \cdot c_{\text{ML}} + w \cdot c_{\text{H}}$.
6. Report **circularity ratio** = $c / 5$. PASS if $c \geq 3.5$ and supporting intact and geometry risk ≤ 0.55 (software gates used for engineering decisions).

Scanning many (x, y) on a fine grid yields a spatial circularity field for fixed EDM settings — the practical realization of the ideation that position matters.

7. Experimental Evidence and Learned Patterns

Graphs below are computed from the sixteen real runs and SEM labels. They justify why the training objective and heuristics look the way they do.

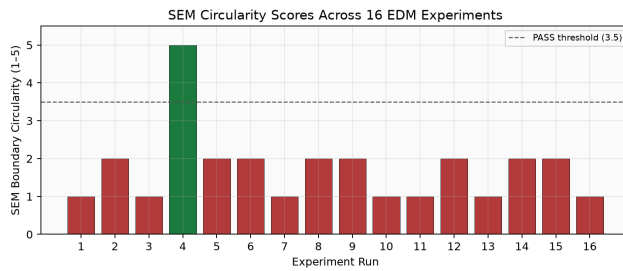


Fig. 5a. SEM circularity by run (only Run 4 passes).

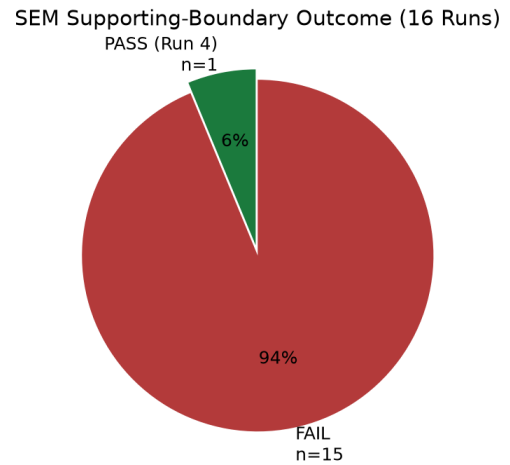


Fig. 5b. 1 PASS / 15 FAIL supporting-ring outcomes.

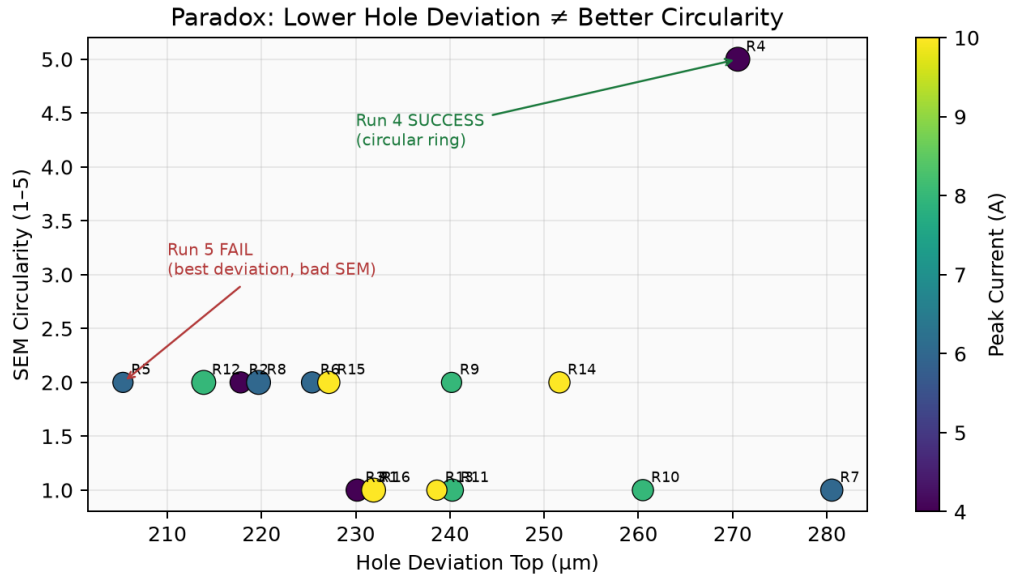


Fig. 6. Why the ML target is SEM circularity: Run 5 (low deviation) fails; Run 4 (higher deviation) is the only circular supporting boundary.

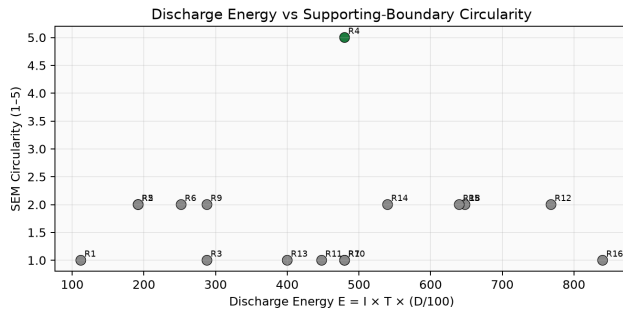


Fig. 7a. Energy $E = I \cdot T \cdot (D/100)$ vs SEM circularity.

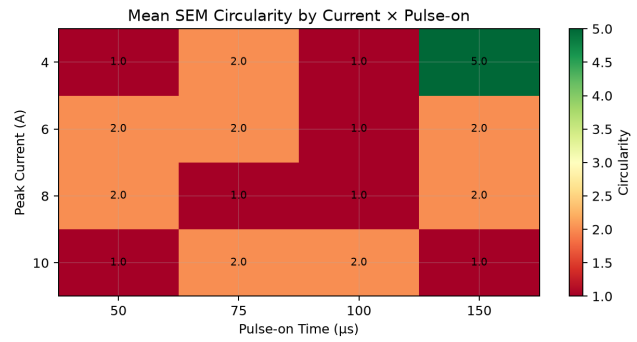


Fig. 7b. Mean circularity: Current × Pulse-on.

The learned neighborhood around Run 4 — low I , long T , high D — matches both SEM truth and GP search recommendations (predicted circularity ≈ 4.6 – 4.9 near 3.6 – 4.8 A / ≈ 150 μ s / ≈ 79 – 80%). Near-strut landings need slightly softer current (≈ 3.5 A) because geometry risk is higher.

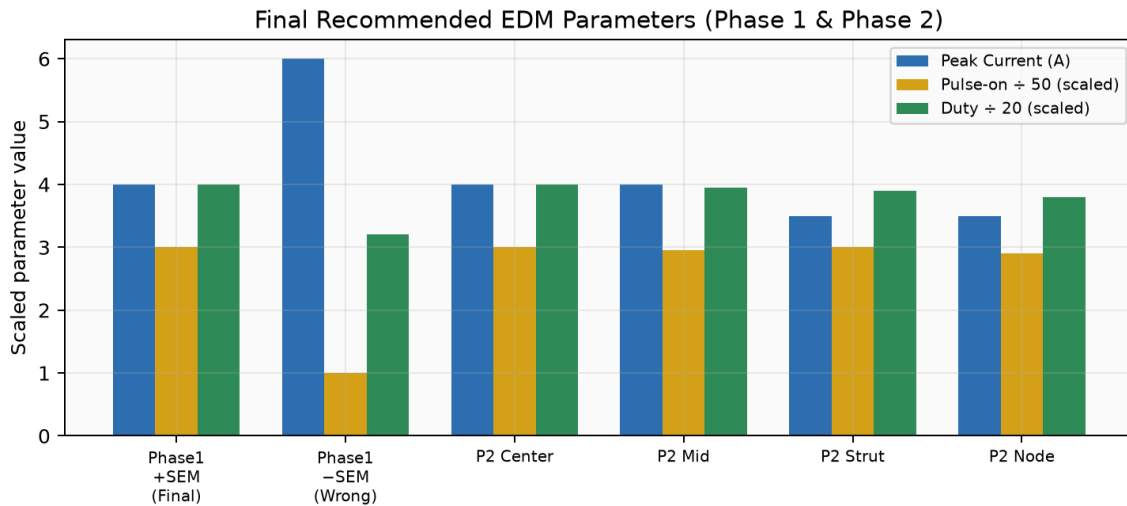


Fig. 8. Final recommended parameter island from the SEM-informed analysis (Phase 1 unknown position; Phase 2 by zone).

8. Final Answers Produced by This Process

Case	Zone	I (A)	T (μ s)	D (%)	Basis
Phase 1	Unknown pos.	4	150	80	SEM Run 4 + GP
Phase 1 reject	Dev-only opt.	6	50	64	Fails SEM ring
Phase 2	Pore center	4	150	80	Low geom. risk
Phase 2	Near strut	3.5	150	78	Higher geom. risk
Phase 2	Near node	3.5	145	76	Corner overlap

Table 3. Final parameter answers from the ML/physics process.

Prediction outputs for any query position: circularity score (1–5), **circularity ratio** = **score/5**, supporting-material integrity, geometry risk, and PASS/FAIL against the thresholds score ≥ 3.5 , ratio ≥ 0.70 , supporting intact, risk ≤ 0.55 .

9. Conclusion

This work shows how a sparse SEM-labeled EDM campaign can still support position-aware circularity prediction. The ideation is simple but decisive: **couple process physics with lattice geometry, expand 16 labels through GP synthesis and risk-modulated spatial replay, train Gradient Boosting, and regularize with heuristics at inference**. That is how circularity ratio becomes computable for any tool position — not by collecting thousands of new SEM trials, but by teaching the model the geometry–process interaction that SEM revealed.

References

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- [4] Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*.
- [5] Lattice Circularity Analyzer Project (2026). Internal modules: `lattice_geometry_engine.py`, `circularity_predictor.py`, `phase1_model_actual.py`; datasets: `original_16_runs.csv`, `run_visual_labels.csv`, `synthetic_1100_points.csv`.